# GLACIS: FIELD GUIDE & PRACTICAL WORKFLOW REFERENCE

**High-Speed Offline Security Assessment Worksheet · eJPTv2 / eCPPT / OSCP Practical Companion**

**What is GLACIS?**
GLACIS is a fast, keyboard-driven terminal worksheet and operational cockpit designed to eliminate cognitive overload during timed practical security assessments. It provides an offline state machine for host discovery, port tracking, credential reuse, syntax playbooks, network pivots, and question proofs. **It does not run autonomous exploits or rely on AI.** You retain 100% human control while GLACIS manages your operational memory.

**INE & OffSec Exam Compliance**
- **100% Passive & Offline:** Verified by AST self-audit (glacis exam-check) — zero network/socket imports.
- **Zero Autonomous Action:** Commands are copied to your clipboard; YOU execute them in your terminal.
- **Zero Cloud / AI Dependencies:** No external API calls, background telemetry, or prohibited LLM endpoints.
- **Permitted Personal Notes:** Functions strictly as an electronic field journal and methodology reference.

## 1. The Six Operational Stations (Switch with Keys 0, 1, 2, 3, 4, 5)

| Station | Name & Purpose | What You See & Do Here | Hotkey |
|---|---|---|---|
| **Station 0** | **Pulse (Triage Board)** | Kill-chain host progression (UNTOUCHED → RECON → FOOTHOLD → USER → ROOT → COMPLETE), progress arithmetic, and explainable next-focus advisories. | `[0]` |
| **Station 1** | **Cockpit (Workbench)** | Targets tree, open ports, service triage status, methodology roadmap, field notes, and interactive command console. | `[1]` |
| **Station 2** | **Playbook Browser** | Browse full tactical playbooks (eJPT workflow, Web App OWASP, Active Directory, Pivoting, Linux & Windows PrivEsc) with instant copy. | `[2]` |
| **Station 3** | **Credential Matrix** | 2D grid of discovered credentials (rows) × target services (cols). Press **[Enter]** to compile spray commands; **[Space]** to cycle verification state. | `[3]` |
| **Station 4** | **Proofs, Loot & Evidence** | Track assessment question answers (:q), user/root flags, disk-backed loot browser ([Space]), screenshot proofing ([v]), and Failure Logs. | `[4]` |
| **Station 5** | **Network & Pivots** | ASCII network topology map of documented subnets, dual-homed pivot gateways, and copy-ready ProxyChains, Chisel, and SSH ProxyJump tunnel actions. | `[5]` |

## 2. Station 0 Pulse: Situation Awareness & Kill-Chain Triage

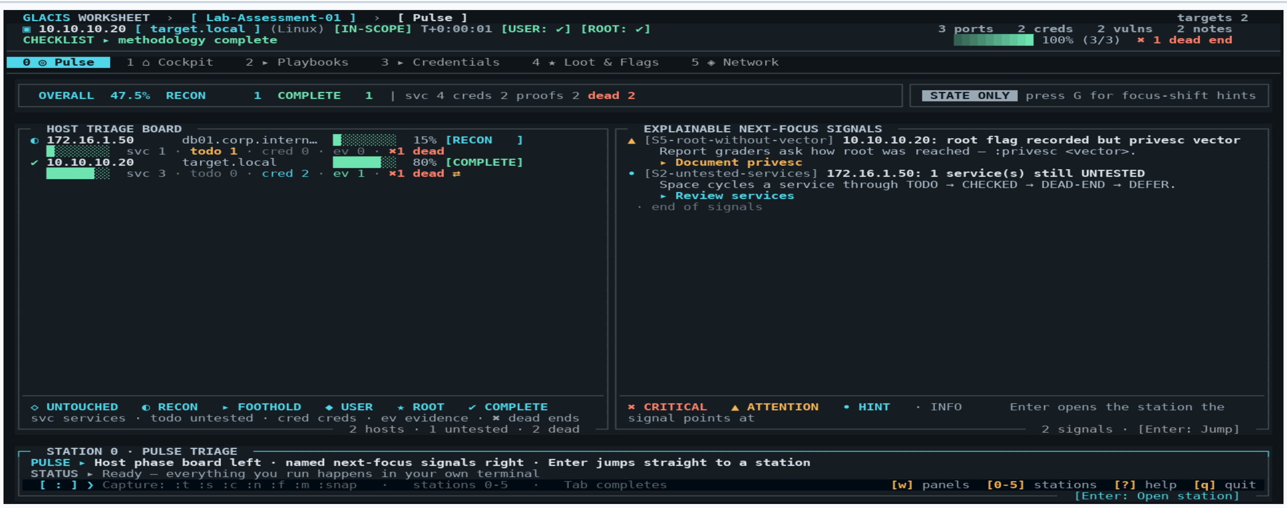


*Figure 1: Station 0 Pulse — Kill-chain progression cards (left), explainable S1-S8 state rules & D1-D4 focus-shift advisories (right), and KPI status band.*

## 3. Station 1 Cockpit Architecture & Real-Time Workspace

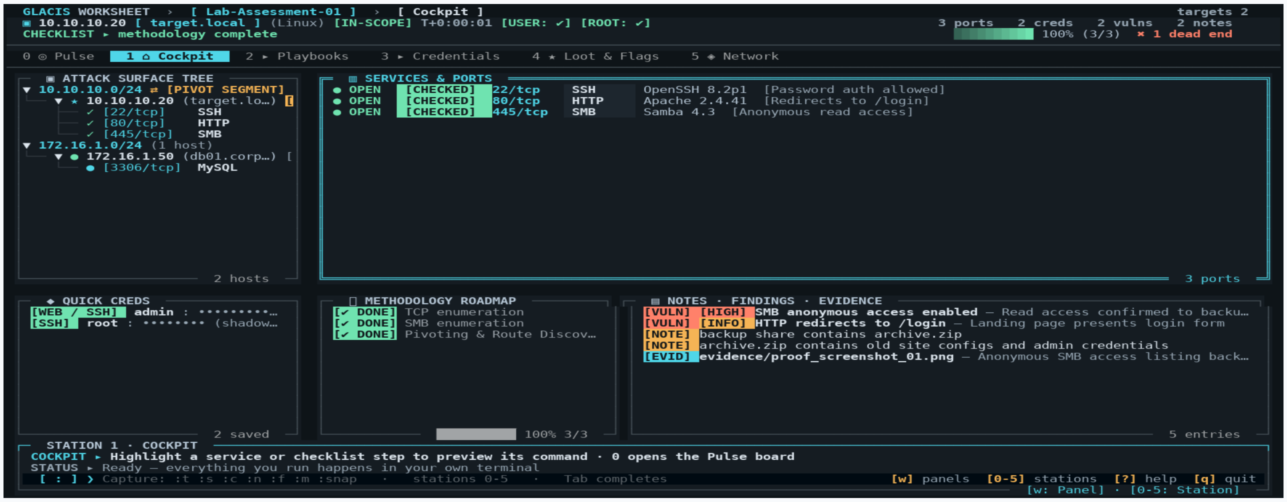


*Figure 2: Station 1 Cockpit — Attack Surface Tree (left), Services & Port Triage (top), Methodology Roadmap (mid), and Bottom Command Console.*

## 4. Keyboard Navigation Matrix ("Muscle Memory Map")

GLACIS is 100% operational from the keyboard. Single-letter keys trigger immediate actions unless a text input is focused:

| Key | Category | Action & Operational Behavior |
|---|---|---|
| `0 – 5` | Station | Switch stations: **[0]** Pulse, **[1]** Cockpit, **[2]** Playbooks, **[3]** Creds Matrix, **[4]** Loot, **[5]** Network. |
| `I (Shift+i)` | Ingestion | **Import Scan & Enum:** Interactive review modal for Nmap (-oX, -oN, -oG) and Web Enum (FFUF, Ferox, Gobuster). Saves evidence to scans/ or enum/. |
| `W (Shift+w)` | Workspace | **Workspace Manager:** Switch between assessment workspaces or scaffold a new per-lab directory (scans/, enum/, screenshots/, loot/). |
| `Tab / Shift+Tab` | Navigation | **Cycle Focus:** Move keyboard focus between panels in Cockpit (double-border indicates focused panel). |
| `j / k (or ↑ / ↓)` | Navigation | **List Navigation:** Move highlight down (j) or up (k) in the active list or tree. |
| `[ / ]` | Targeting | **Target Switcher:** Switch immediately to previous [ or next ] target machine. |
| `: (colon)` | Console | **Focus Command Bar:** Instantly activates the bottom console input ready to type commands. |
| `Esc` | Console | **Dismiss / Blur:** Clears command input and returns focus directly to the active workbench. |
| `Space` | Triage / State | • **On Service:** Cycle status: UNTESTED → [CHECKED] → [DEAD-END] → [DEFERRED].<br>• **In Station 3 Matrix:** Cycle test state: [UNTESTED] → [VALID] → [PWN3D] → [INVALID].<br>• **In Station 4 Loot:** Open **Loot Preview Modal** to inspect hashes, configs, or keys. |
| `, / .` | Tool Carousel | **Recipe Carousel:** When a service is highlighted, press **.** (next) or **,** (prev) to cycle alternative tool recipes in the console. |
| `Enter` | Execute / Copy | • **On Service / Recipe:** Copies the previewed tool command to clipboard.<br>• **In Station 3 Matrix:** Compiles & copies ready-to-run spray command for credential & port.<br>• **In Station 5 Network:** Copies selected pivot/tunnel syntax to clipboard.<br>• **In Station 0 Pulse:** Jumps directly to the station that owns the highlighted advisory. |
| `v` | Evidence | **Paste Screenshot:** In Station 4, saves clipboard image directly to screenshots/ and links as Evidence. |
| `y` | Quick Copy | Copies selected entity's primary value (IP address, port, password, or checklist command). |
| `z` | Layout | **Zoom:** Maximize the focused panel to full-screen; press **z** again to restore normal layout. |
| `T (Shift+t)` | Theme | **Theme Picker:** Open theme modal with 8 WCAG AAA palettes (1-8 keys, **d** sets default). |
| `G (Shift+g)` | Guidance | **Toggle Guidance:** Enables or disables opt-in focus-shift advisories (D-rules) on Station 0. |
| `/ or Ctrl+F` | Search | **Global Fuzzy Search:** Search across targets, services, credentials, checklists, and notes. |
| `r` | Reference | **Quick Reference Modal:** Open offline practical reference for instant syntax lookup. |
| `t / s / c / n` | Fast Capture | Modal dialogs to quickly add Target (t), Service (s), Credential (c), or Note (n). |
| `a` | Question Proof | In Station 4: Open **Add Proof** modal to link question/item number to proof evidence. |

## 5. Dual-Terminal Operating Model & Universal CLI Ingestion

The standard professional workflow runs GLACIS TUI in one terminal pane while executing active scans and exploits in an adjacent shell. Because GLACIS uses a shared SQLite store with WAL mode, changes committed in your attack terminal update the TUI live.

| Workflow & Goal | Command Line Execution (In Attack Terminal) | Operational Result in GLACIS |
|---|---|---|
| **Automated Scan Ingest** | nmap -sC -sV $IP -oX scans/box.xml &&<br>glacis import --apply scans/box.xml | Parses all hosts, ports, services, and banners into SQLite. Raw XML archived into scans/ as immutable evidence. |
| **1-Liner Shell Wrapper** | gnmap() { nmap -sC -sV "$@" -oX scans/s.xml &&<br>glacis import --apply scans/s.xml; } | Type gnmap 10.10.10.20: scans and populates GLACIS with zero manual typing. |
| **Direct Terminal Pipe** | nikto -h $IP \| glacis extract - --apply<br>cat enum.log \| glacis extract - --apply | Parses raw terminal text on the fly, extracting targets, ports, hashes, and CTF flags. |
| **Target & Service Add** | glacis target 10.10.10.20 --os Linux<br>glacis service 10.10.10.20 80/tcp http | Direct CLI insertion from background scripts or terminal loops without opening the TUI. |
| **Discovered Credential** | glacis cred admin:Password123! -t 10.10.10.20 | Immediately adds credentials into the Station 3 Credential Matrix. |
| **Flag & Proof Capture** | glacis proof -t 10.10.10.20 --flag user "eJPT{...}"<br>glacis proof --q-num Q7 --proof "admin:P@ss" | Records objective flags and exam question proofs directly into the Station 4 ledger. |
| **Snapshot Safety Net** | glacis snapshot create -m "pre-privesc"<br>glacis snapshot list / glacis snapshot restore <file> | Creates byte-consistent online SQLite backup. Rotates automatically (keeps newest 5 snapshots). |
| **Offline Compliance Audit** | glacis exam-check | AST static scan proving zero network/socket imports across all 38 codebase modules. |

## 6. Quick Command Console Syntax (Press [:] Anywhere in TUI)

| Console Command | Description & Execution | Syntax Example |
|---|---|---|
| :0 – :5 | Switch stations: :0 Pulse, :1 Cockpit, :2 Playbooks, :3 Creds, :4 Loot, :5 Network. | :0 :pulse :network |
| :t <ip> | Quickly add new target host into active workspace. | :t 10.10.10.20 |
| :s <port> <svc> | Add open port/service under currently selected target. | :s 445/tcp smb :s 8080 http |
| :c <user:pass> | Add credential pair into Vault and auto-mount to Spray Matrix. | :c admin:Summer2024! |
| :n / :f | Record field note (:n) or security finding (:f). | :n backup dir found<br>:f anonymous smb |
| :uflag / :rflag | Record captured user flag or root flag hash. | :uflag eJPT{user_hash} |
| :pivot <route> | Mark active target as dual-homed gateway; updates Station 5 topology. | :pivot 172.16.1.0/24 via socks5:1080 |
| :snap [note] | Take an online byte-consistent database snapshot safety net. | :snap before-bruteforce |
| :stuck / :clue | Log rabbit-hole stuck point or breakthrough clue in Failure Log. | :stuck hydra lockout<br>:clue check backup.zip |
| :q <num> <proof> | Pin exam question answer proof (e.g. hash, flag, config path). | :q 14 root:$6$hash... |
| :export html | Generate standalone, printable HTML handover report (zero JS/CDN). | :export html exam_final.html |
| :export exam | Export exam evidence bundle to exam_evidence.md. | :export exam |
| :export wordlists | Harvest all discovered credentials into loot/users.txt and loot/passwords.txt. | :export wordlists |
| :theme <name> | Switch visual palette (slate, midnight, ember, cyber, sugary, candy, caramel, catppuccin). | :theme midnight :theme cyber |

## 7. Step-by-Step Practical Assessment Workflow ("The Battle Plan")

Follow this battle-tested loop for each target in practical exams (eJPTv2, eCPPT, OSCP) to maintain 100% rigor and never lose progress:

| Assessment Stage | Step-by-Step Operator Action & Commands |
|---|---|
| **Phase 1: Workspace Init & Scoping** | 1. Create dedicated lab directory: glacis init lab_name --ip 10.10.10.20.<br>2. Set attacker VPN IP: :lhost auto (or :lhost 192.168.0.50) and :lport 4444.<br>3. Apply exam methodology checklist: :m ejpt in Cockpit. |
| **Phase 2: Discovery & Ingestion** | 1. Run port scan in attack terminal: nmap -sC -sV $IP -oX scans/init.xml.<br>2. Ingest automatically: glacis import --apply scans/init.xml (or press **[I]** in TUI).<br>3. Verify Station 0 Pulse reflects attack surface: host moves from **UNTOUCHED** to **RECON**. |
| **Phase 3: Port Triage & Tool Recipes** | 1. Highlight open port in Cockpit with j / k.<br>2. Cycle alternative tool recipes in console with . (e.g. feroxbuster → nikto → curl).<br>3. Press **[Enter]** to copy recipe to clipboard; run in terminal.<br>4. Press **[Space]** on service to mark [CHECKED] or [DEAD-END]. |
| **Phase 4: Credential Harvest & Spray Validation** | 1. Record found credentials: :c admin:Summer2024!.<br>2. Switch to Station 3 (press **[3]**) to inspect the 2D Credential Spray Matrix.<br>3. Press **[Enter]** on any matrix cell to copy ready-to-run spray command.<br>4. Press **[Space]** on cell to record outcome: [VALID] or [PWN3D].<br>5. Harvest discovered accounts: :export wordlists writes loot/users.txt. |
| **Phase 5: Foothold, Flags & Anti-Rabbit Hole** | 1. Record initial shell foothold: :foothold php-reverse-shell.<br>2. Capture user flag: :uflag eJPT{...}.<br>3. If stuck for >20 mins, avoid rabbit hole: log dead-end with :stuck <why> and check Station 0 Pulse for next un-checked surface.<br>4. When progress is made, log breakthrough: :clue <what_worked>. |
| **Phase 6: Internal Pivoting & Subnet Expansion** | 1. Discovered internal route? Mark target dual-homed: :pivot 172.16.1.0/24 via socks5:1080.<br>2. Switch to Station 5 (press **[5]**) to inspect ASCII network graph.<br>3. Press **[Enter]** on actions to copy ProxyChains config, Chisel server/client, or SSH ProxyJump syntax.<br>4. Add pivot target: :t 172.16.1.50 and begin Phase 2 scan through proxy. |
| **Phase 7: PrivEsc & Final Submission Handover** | 1. Switch to Station 2 (press **[2]**) and search Linux / Windows PrivEsc playbooks.<br>2. Record root flag: :rflag eJPT{...} and document privesc: :privesc SUID-pkexec.<br>3. Log question proofs: :q 14 /etc/shadow-hash (press **[a]** in Station 4).<br>4. Export standalone printable HTML handover: :export html final_handover.html.<br>5. Export evidence markdown bundle: :export exam writes exam_evidence.md. |

**SAFETY NET: AUTOMATIC & ON-DEMAND SNAPSHOTS (:snap)**
Lab VMs reset frequently, and databases can corrupt during power drops. GLACIS automatically creates a snapshot before scan imports. Take on-demand snapshots with :snap before-pivot or glacis snapshot create. Restore losslessly at any time with glacis snapshot restore <file>. Snapshots use SQLite's native backup API and keep the newest 5 copies automatically rotated.

## 8. Station 3 Deep Dive: 2D Credential Spray Matrix

Station 3 eliminates credential amnesia by organizing all discovered credentials (rows) against all target authentication services (columns) in a live 2D grid:

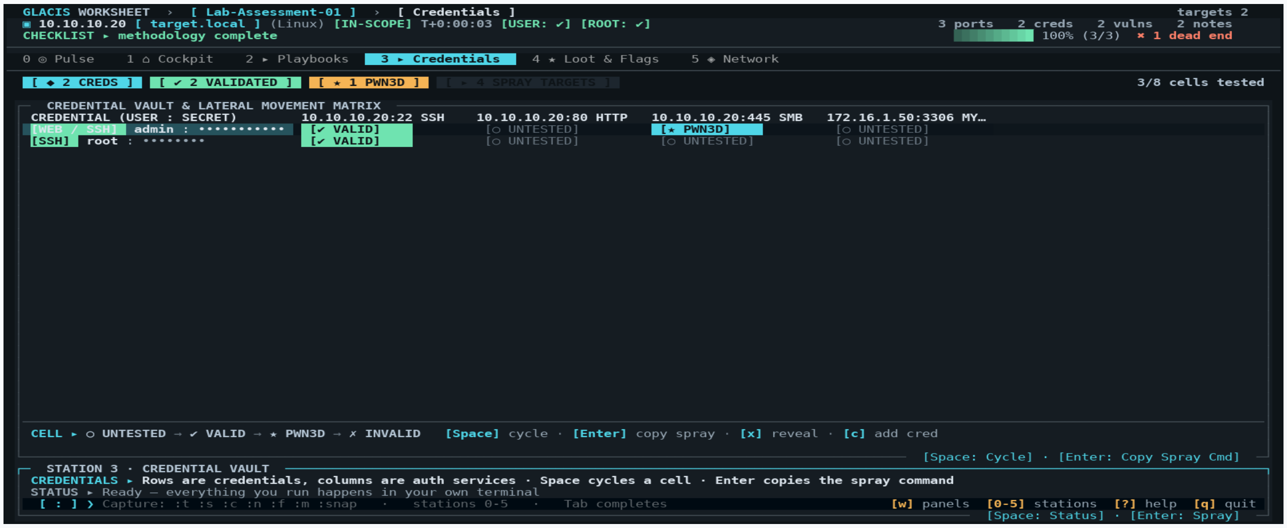


*Figure 3: Station 3 Credential Matrix — 2D grid of discovered accounts against authenticating services. Enter copies spray command; Space cycles verification state.*

| Matrix Element | Hotkey & Action | Operational Impact |
|---|---|---|
| **Cell Status Toggle** | `Press [Space]` | Cycles cell through [UNTESTED] → [VALID] (green) → [PWN3D] (magenta) → [INVALID] (red). |
| **Auto-Compile Spray Syntax** | `Press [Enter] on cell` | Compiles ready-to-run NetExec, Hydra, SSH, or Evil-WinRM command for that specific user, password, target IP, and port. |
| **Credential Vault Table** | Left Panel (navigate with **h / l**) | Inventory of all discovered accounts, source tracking (e.g. shadow crack, wp-config, SAM dump), and validation flags. |
| **Harvest Wordlists** | `:export wordlists` | Exports all captured usernames to loot/users.txt and passwords to loot/passwords.txt for instant feeding to wordlist attacks. |

## 9. Station 4 Deep Dive: Question Proofs & Loot Ledger

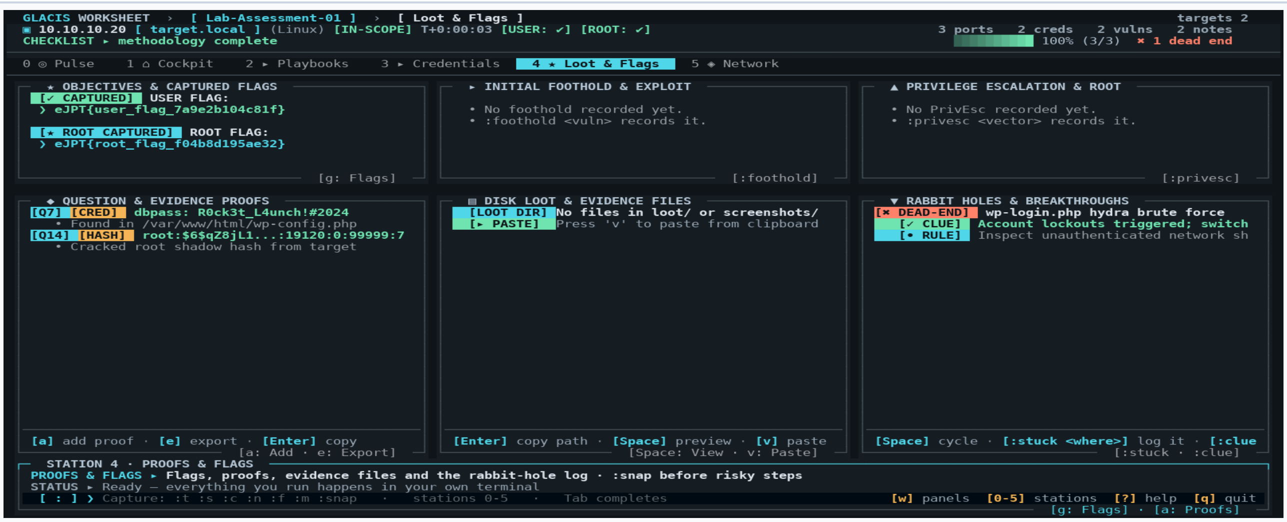


*Figure 4: Station 4 Proofs & Loot Ledger — Question Proofs, User/Root Flags, Disk Loot Browser, and Anti-Rabbit Hole Failure Log.*

## 10. Station 5 Deep Dive: Network Topology, Dual-Homed Pivots & Tunnels

Station 5 visualizes internal network architecture and dual-homed pivot gateways directly from your documented target IPs and :pivot notes:

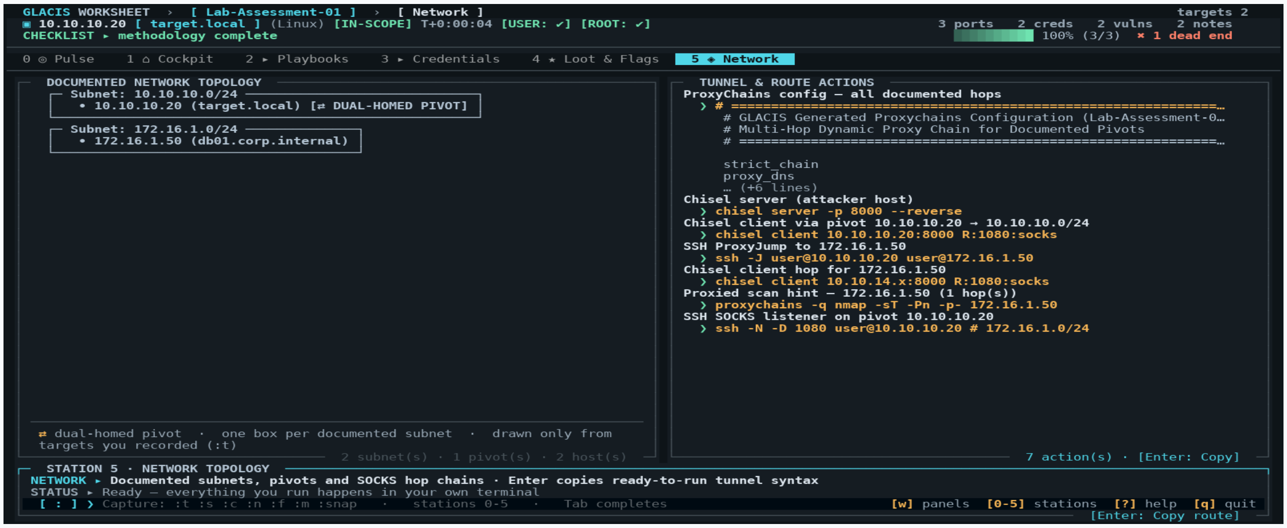


*Figure 5: Station 5 Network — Documented ASCII Subnet Topology (left) and Copy-Ready Tunnel & Routing Actions (right). Enter copies syntax to clipboard.*

| Feature / Capability | Command & Hotkey | Operational Purpose |
| --- | --- | --- |
| **Mark Dual-Homed Pivot** | `:pivot <route>` (e.g. `:pivot 172.16.1.0/24 via socks5:1080`) | Registers host as pivot gateway between subnets. The ASCII topology graph draws the cross-subnet connection automatically. |
| **ProxyChains Config** | Highlight action & press **[Enter]** | Copies complete proxychains4.conf snippet configured for your active SOCKS proxy port. |
| **Chisel Tunnel Recipes** | Highlight action & press **[Enter]** | Copies ready-to-run Chisel server command (attacker) and Chisel client reverse tunnel command (victim). |
| **SSH Dynamic Forwarding** | Highlight action & press **[Enter]** | Copies ssh -D 1080 -N -f user@pivot_ip or SSH ProxyJump commands with target IP and credentials populated. |

## 11. Station 2 Deep Dive: Playbook Browser & Methodology Roadmaps

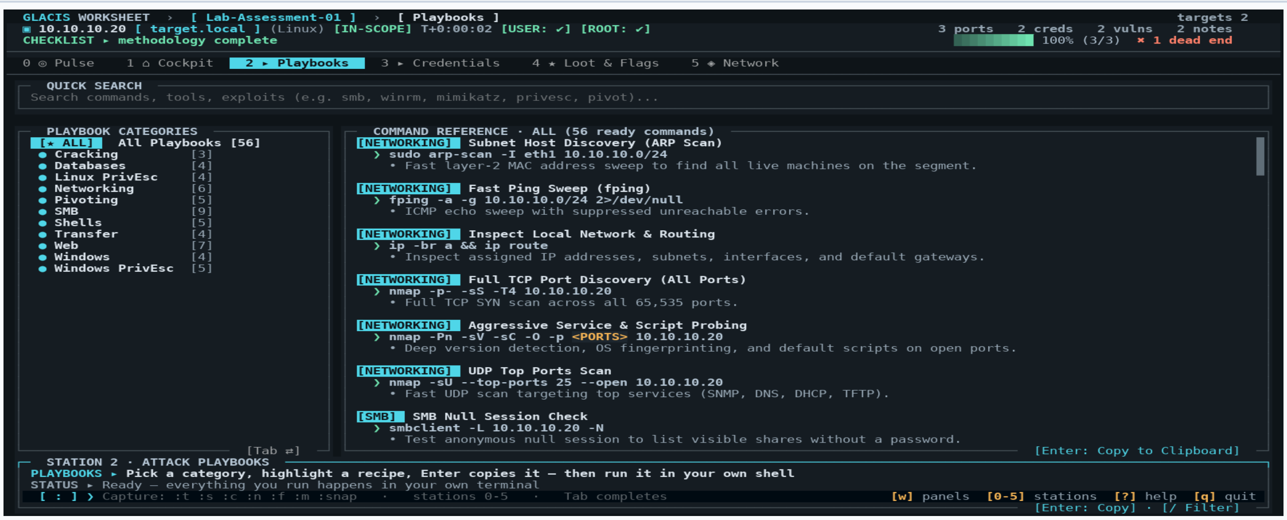


*Figure 6: Station 2 Playbooks — Categorized command encyclopedia and multi-tool recipe carousel. Enter copies recipe to clipboard.*

## 12. Standalone HTML Handover Reports & Evidence Bundles

At the end of an assessment, or before finalizing answers in the exam portal, export self-contained submission dossiers:

| Export Format | Command Line & Console | Deliverable & Features |
|---|---|---|
| **Standalone HTML Report** | `glacis export -f html -o report.html`<br>`Console: :export html` | Single self-contained HTML file. **Zero JavaScript, zero external CDN/fonts.** Includes engagement overview, target dossiers, masked creds, network topology, and @media print stylesheet for clean PDF printout. |
| **Unmasked HTML Report** | `glacis export -f html -o report.html --reveal-creds` | Reveals all plaintext passwords for internal auditor review or personal grading. |
| **Markdown Evidence Bundle** | `glacis export -f md -o exam.md`<br>`Console: :export exam` | Generates exam_evidence.md with all recorded question proofs (Q1-Q35), hashes, flags, and golden reproduction commands. |
| **Lossless JSON Backup** | `glacis export -f json -o backup.json` | Complete structured export of all database tables for archival or migration. |

## 13. Built-In Methodology Templates (Press [m] in Cockpit)

| Template Alias | Category | Key Checklist Steps & Focus |
|---|---|---|
| `:m ejpt` | eJPT Practical | Scope recon → Host discovery → Full TCP scan → Service triage → Low-hanging fruit → Web fuzzing → Cred spray → Pivoting → PrivEsc. |
| `:m web` | OWASP Web App | Technology profiling (whatweb) → Directory fuzzing (feroxbuster) → Parameter fuzzing → SQLi → Auth bypass → LFI/RFI → File upload. |
| `:m smb` | SMB & Windows | Null session shares (smbclient/smbmap) → User enum (rpcclient) → Password policy → Anonymous signing audit → Vulnerability scan (MS17-010). |
| `:m pivoting` | Network Pivoting | Dual-homed adapter discovery (ip a) → Internal route inspect → Proxychains & Chisel server setup → Port forward (socat) → Subnet sweep. |
| `:m privesc_linux` | Linux PrivEsc | sudo -l commands → SUID/SGID binaries (GTFOBins) → Capabilities (getcap) → World-writable scripts & crontabs → Internal listening ports. |
| `:m privesc_windows` | Windows PrivEsc | whoami /priv (SeImpersonate) → Unquoted service paths → Modifiable service binaries → Stored credentials (cmdkey) → Registry backup. |

## 14. The Eight WCAG AAA Palettes (Press [T] to Switch)

- **Slate** (Default): Glacial cyan accent on deep arctic granite. Clean, professional, high-contrast.
- **Midnight:** Indigo and periwinkle on deep nocturne navy. Easy on the eyes during late-night assessments.
- **Ember:** Amber CRT glow on warm dark charcoal. Classic retro phosphor terminal aesthetic.
- **Cyber:** Electric cyan and hot pink on pitch black. High energy modern Tokyo nightscape.
- **Sugary:** Soft espresso ink on unbleached vanilla cream. Soothing warm daylight mode.
- **Candy:** Deep violet on soft cotton lilac. Playful pastel with verified 7:1 contrast.
- **Caramel:** Teal instrument accent on toffee parchment. High-legibility technical document tone.
- **Catppuccin:** Sapphire and lavender on warm mocha. The beloved community dark palette.

**EXAM / ASSESSMENT DAY QUICK START:**

1. **Verify Offline Compliance:** Run glacis exam-check in terminal (proves zero network imports for AI proctor).
2. **Launch GLACIS:** Open terminal and launch glacis (or short alias gls).
3. **Scaffold Dedicated Workspace:** Run :init target_lab <ip> to create a fresh directory with scans/, enum/, screenshots/, and loot/.
4. **Set VPN & Methodology:** Run :lhost auto and apply template with :m ejpt.
5. **Live Triage:** Use Station 0 (**[0]**) to track kill-chain progress; use Station 5 (**[5]**) when internal pivots are discovered.
6. **Handover Dossier:** Run :export html or :export exam before final submission to verify all answers.